Phase1: Understanding the NHANES 2021–2023 Dataset

"Evaluating the Association of Cigarette Smoking and Physical Activity with Multiple Metabolic(BMI and Blood pressure) and Lipid Biomarkers (Glycohemoglobin, Fasting Glucose, HDL-Cholesterol, Total Cholesterol, and Triglycerides) Among Adults(Males and Females, Aged 18 and Older) in NHANES"

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1 Data Source and Data Collection

1.1 Source of the Data

This project uses secondary data from the National Health and Nutrition Examination Survey (NHANES), specifically the August 2021–August 2023 data cycle. NHANES is conducted by the National Center for Health Statistics (NCHS), which is part of the U.S. Centers for Disease Control and Prevention (CDC). Its purpose is to assess the health and nutritional status of adults and children in the United States by combining questionnaire interviews, physical examinations, and laboratory measurements.

The NHANES is a cross-sectional observational study that collects data through standardized questionnaires, physical examinations, and laboratory measurements.

1.2 Limitation Due COVID-19 Pandemic

Sample selection within dwelling units during the August 2021–August 2023 cycle differed from NHANES befor cycles,due to the COVID-19 pandemic, in two important ways:

- **Elimination of oversampling by race, Hispanic origin, and income:**

  To minimize screening visits, oversampling by race, Hispanic origin, and income was completely eliminated. Dwelling units were no longer preassigned demographic criteria. Instead, at least one resident in every sampled dwelling unit was selected as eligible, provided that the household met the basic occupancy requirements.

- **Modification of person-level age oversampling:**

  Age-based oversampling procedures were modified so that all individuals aged 0–19 years and 60 years or older encountered in a household were selected. Selection among adults aged 20–59 years depended on household size:

  - In households with 1–3 adults, one adult was randomly selected.
  - In households with 4 or more adults, two adults were randomly selected.

2 Observational Units and Data Structure

The observational unit is an individual NHANES participant. Each row in the merged analytical dataset represents one participant, rather than one household, one examination, or one blood sample.

For instance, the Body Measurements dataset contains a comprehensive set of anthropometric measurements collected from NHANES participants. For the purposes of the present study, only the Body Mass Index (BMI) variable was extracted from this dataset, while the remaining measurements were excluded from the analysis.

For transparency and reproducibility, a complete list of the variables included in the study, along with their descriptions, is provided below:

Table 1: NHANES variables used in the analysis.

Variable Name	**Variable Description**	**Data File Name**	**Data File Description**	**Component**
SEQN	Respondent sequence number (unique participant identifier)	DEMO_L	Demographic Variables and Sample Weights	Demographics
BMXBMI	Body Mass Index (kg/m^2)	BMX_L	Body Measures	Examination
BPXODI1	Diastolic – 1st oscillometric reading	BPXO_L	Blood Pressure – Oscillometric Measurements	Examination
BPXODI2	Diastolic – 2nd oscillometric reading	BPXO_L	Blood Pressure – Oscillometric Measurements	Examination
BPXODI3	Diastolic – 3rd oscillometric reading	BPXO_L	Blood Pressure – Oscillometric Measurements	Examination
BPXOSY1	Systolic – 1st oscillometric reading	BPXO_L	Blood Pressure – Oscillometric Measurements	Examination
BPXOSY2	Systolic – 2nd oscillometric reading	BPXO_L	Blood Pressure – Oscillometric Measurements	Examination
BPXOSY3	Systolic – 3rd oscillometric reading	BPXO_L	Blood Pressure – Oscillometric Measurements	Examination
LBXGH	Glycohemoglobin (%)	GHB_L	Glycohemoglobin	Laboratory
LBXGLU	Fasting Glucose (mg/dL)	GLU_L	Plasma Fasting Glucose	Laboratory
LBDHDD	Direct HDL-Cholesterol (mg/dL)	HDL_L	Cholesterol – High-Density Lipoprotein	Laboratory
LBXTC	Total Cholesterol (mg/dL)	TCHOL_L	Cholesterol – Total	Laboratory
LBXTLG	Triglyceride (mg/dL)	TRIGLY_L	Cholesterol – Low-Density Lipoproteins (LDL) & Triglycerides	Laboratory
PAD790Q	How often does the participant engage in moderate-intensity leisure-time physical activities?	PAQ_L	Physical Activity	Questionnaire
PAD790U	Unit of frequency for moderate-intensity leisure-time physical activities	PAQ_L	Physical Activity	Questionnaire
PAD800	Duration of moderate-intensity leisure-time physical activities per session	PAQ_L	Physical Activity	Questionnaire
PAD810Q	How often does the participant engage in vigorous-intensity leisure-time physical activities?	PAQ_L	Physical Activity	Questionnaire

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Variable Name	Variable Description	Data File Name	Data File Description	Component
PAD810U	Unit of frequency for vigorous-intensity leisure-time physical activities	PAQ_L	Physical Activity	Questionnaire
PAD820	Duration of vigorous-intensity leisure-time physical activities per session	PAQ_L	Physical Activity	Questionnaire
SMD650	Number of cigarettes smoked per day on smoking days during the past 30 days	SMQ_L	Smoking – Cigarette Use	Questionnaire
SMQ020	Whether the participant has smoked at least 100 cigarettes in their lifetime	SMQ_L	Smoking – Cigarette Use	Questionnaire
SMQ040	{Do you/Does SP} now smoke cigarettes?	SMQ_L	Smoking – Cigarette Use	Questionnaire
RIDAGEYR	Age in years at screening; participants aged 80 years and older are top-coded at 80	DEMO_L	Demographic Variables and Sample Weights	Demographics
INDFMPIR	Ratio of family income to poverty	DEMO_L	Demographic Variables and Sample Weights	Demographics
RIAGENDR	Gender of the participant	DEMO_L	Demographic Variables and Sample Weights	Demographics
RIDEXPRG	Pregnancy status for females aged 20–44 years at the time of the MEC examination	DEMO_L	Demographic Variables and Sample Weights	Demographics
WTINT2YR	Full sample 2-year interview weight	DEMO_L	Demographic Variables and Sample Weights	Demographics
WTMEC2YR	Full sample 2-year MEC examination weight	DEMO_L	Demographic Variables and Sample Weights	Demographics
SDMVPSU	Masked variance unit pseudo-PSU variable for variance estimation	DEMO_L	Demographic Variables and Sample Weights	Demographics
SDMVSTRA	Masked variance unit pseudo-stratum variable for variance estimation	DEMO_L	Demographic Variables and Sample Weights	Demographics
WTSAF2YR	Fasting Subsample 2 Year MEC Weight	GLU_L, TRIGLY_L	Plasma Fasting Glucose; Cholesterol – Low-Density Lipoproteins (LDL) & Triglycerides	Laboratory
WTPH2YR	Phlebotomy 2 Year Weight	GHB_L, HDL_L, TCHOL_L	Glycohemoglobin; Cholesterol – High-Density Lipoprotein; Cholesterol – Total	Laboratory

The participant is identified by the variable `SEQN`, which is the respondent sequence number. This identifier is used to combine information about the same participant across the different NHANES component files.

Finally, the raw NHANES datasets were processed and organized using Python. The relevant variables were extracted from their respective source files and combined with the corresponding participant identifiers, sampling weights, and survey control variables, where applicable. Separate analytical datasets were subsequently prepared for each variable of interest to facilitate data processing and statistical analysis.

3 Sampling and Data Structure

3.1 Collection Data Method

Initially, a four-year sample was designed for the period 2019–2022. However, data collection was discontinued in March 2020 due to the COVID-19 pandemic. When SARS-CoV-2 infections had subsided sufficiently to resume data collection, the decision was made not to collect data from the remainder of the 2019–2022 sample. Instead, a new sample was drawn with the aim of limiting contact between NHANES staff and participants.

NHANES resumed data collection in August 2021 using a new two-year sample design similar to previous designs, which employed a multi-year, stratified, clustered, four-stage probability sample of the U.S. population.

According to the CDC report NHANES 2017March2020 Methodology Report, the data collection methodology employs a multifaceted approach designed to produce nationally representative estimates of the noninstitutionalized civilian population of the United States.

The primary data collection methodology consists of the following components,

- **Complex Four-Stage Sample Design**

  NHANES uses a stratified, multistage probability sample design consisting of four stages:

  1. **Stage 1:** Selection of primary sampling units (PSUs), which are typically counties or groups of counties across the United States.
  2. **Stage 2:** Selection of segments, such as city blocks or combinations of blocks, within the selected PSUs.
  3. **Stage 3:** Selection of specific households within the selected segments.
  4. **Stage 4:** Selection of individuals within sampled households based on demographic characteristics such as age, sex, race, and ethnicity.

- **Personal Interviews**

  Participants complete comprehensive in-person interviews, often conducted using Computer-Assisted Personal Interviewing (CAPI) systems. Trained interviewers collect detailed demographic, socioeconomic, dietary, and health-related information.

- **Standardized Physical Examinations**

  Health and nutritional data are collected through standardized physical examinations conducted in specially designed and equipped Mobile Examination Centers (MECs). These centers travel to selected study locations across the country.

  The examinations include medical and dental assessments, physiological measurements, and laboratory tests performed on biological specimens, such as blood and urine.

- **Estimation and Weighting Procedures**

  Because NHANES employs a complex probability sample, sample weights are calculated and applied during data analysis. These weights account for unequal probabilities of selection, nonresponse adjustments, and differences between the sample and the target U.S. population.

Consequently, the resulting estimates are designed to be unbiased and nationally representative.

As discussed in Section 1, several differences exist between NHANES 2017March 2020 and NHANES 20212023, which may affect the sampling weights required for statistical analysis. These differences must be taken into account when selecting and applying the appropriate sampling weights in the subsequent analysis.

The main analytical population for this project is adults aged 20 years and older. This restriction is applied using `RIDAGEYR`, the participant's age in years. The final number of observations may be smaller than the number of eligible participants because not every participant has a valid response or measurement for every selected variable.

3.2 Sampling and Generalizability

NHANES is an observational study based on a complex, multistage probability sampling design. The data collection for NHANES 20212023 was conducted using a multiyear, stratified, clustered, four-stage sample of the U.S. civilian noninstitutionalized population. Accordingly, NHANES provides a nationally representative sample of the target population when appropriate sampling weights and survey design variables are incorporated into statistical analyses.

Sampling weights are assigned to participants to account for unequal probabilities of selection, survey nonresponse, and adjustments for population coverage. These weights allow individual observations to contribute appropriately to population-level estimates, thereby improving the representativeness of statistical results.

Different sampling weights are provided depending on the survey component from which the variables were collected. The appropriate weight must be selected according to the variables included in each analysis, following the principle of using the weight corresponding to the smallest applicable subsample.

For the NHANES 20212023 dataset, the principal sampling weights used in this study include:

- **WTINT2YR:** Full-sample two-year interview weight, applicable to variables collected during the household interview.
- **WTMEC2YR:** Full-sample two-year Mobile Examination Center (MEC) examination weight, applicable to variables collected during the MEC examination.
- **WTSAF2YR:** Two-year fasting subsample weight, applicable to laboratory measurements requiring fasting, such as fasting plasma glucose and triglycerides.
- **WTPH2YR:** Two-year phlebotomy weight, applicable to laboratory analyses based on blood specimens, accounting for additional nonresponse associated with blood collection.

In addition to sampling weights, NHANES provides two survey design variables for variance estimation:

- **SDMVPSU:** Masked variance-unit pseudo-primary sampling unit (pseudo-PSU) identifier.

- **SDMVSTRA:** Masked variance-unit pseudo-stratum identifier.

These variables are used together with the appropriate sampling weights to account for the clustered and stratified sampling design when estimating sampling variances, standard errors, confidence intervals, and conducting statistical hypothesis tests.

Consequently, both the sampling weights and the survey design variables must be appropriately incorporated into the statistical analysis to obtain valid population estimates and reliable measures of statistical uncertainty.

4 Variable Classification

Variables in this project are classified according to their analytical role and their measurement scale. A variable may be an outcome, an explanatory variable, or a control variable. Separately, it may be quantitative or categorical.

Table 2: Variables included in the study and their classifications.

Variable Name	**Variable Description**	**Measurement Scale**	**Data Classification**
BMXBMI	Body Mass Index (kg/m^2)	Continuous quantitative	Response
BPXODI1	Diastolic blood pressure – 1st oscillometric reading	Continuous quantitative	Response
BPXODI2	Diastolic blood pressure – 2nd oscillometric reading	Continuous quantitative	Response
BPXODI3	Diastolic blood pressure – 3rd oscillometric reading	Continuous quantitative	Response
BPXOSY1	Systolic blood pressure – 1st oscillometric reading	Continuous quantitative	Response
BPXOSY2	Systolic blood pressure – 2nd oscillometric reading	Continuous quantitative	Response
BPXOSY3	Systolic blood pressure – 3rd oscillometric reading	Continuous quantitative	Response
LBXGH	Glycohemoglobin (%)	Continuous quantitative	Response
LBXGLU	Fasting glucose (mg/dL)	Continuous quantitative	Response
LBDHDD	Direct HDL cholesterol (mg/dL)	Continuous quantitative	Response
LBXTC	Total cholesterol (mg/dL)	Continuous quantitative	Response

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Variable Name	Variable Description	Measurement Scale	Data Classification
LBXTLG	Triglycerides (mg/dL)	Continuous quantitative	Response
PAD790Q	Frequency of moderate-intensity leisure-time physical activities	Discrete quantitative	Primary Explanatory
PAD790U	Unit of frequency for moderate-intensity leisure-time physical activities	Nominal categorical	Primary Explanatory
PAD800	Duration of moderate-intensity leisure-time physical activities per session	Discrete quantitative	Primary Explanatory
PAD810Q	Frequency of vigorous-intensity leisure-time physical activities	Discrete quantitative	Primary Explanatory
PAD810U	Unit of frequency for vigorous-intensity leisure-time physical activities	Nominal categorical	Primary Explanatory
PAD820	Duration of vigorous-intensity leisure-time physical activities per session	Discrete quantitative	Primary Explanatory
SMD630	Age at which the participant first smoked a whole cigarette	Discrete quantitative	Primary Explanatory
SMD650	Number of cigarettes smoked per day on smoking days during the past 30 days	Discrete quantitative	Primary Explanatory
SMQ020	Whether the participant has smoked at least 100 cigarettes in their lifetime	Binary categorical	Primary Explanatory
SMD641	Number of days the participant smoked a cigarette during the past 30 days	Discrete quantitative	Primary Explanatory
RIDAGEYR	Age in years at screening; participants aged 80 years and older are top-coded at 80	Discrete quantitative	Control
INDFMPIR	Ratio of family income to poverty	Continuous quantitative	Control
RIAGENDR	Sex of the participant	Nominal categorical	Control

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Variable Name	Variable Description	Measurement Scale	Data Classification
RIDEXPRG	Pregnancy status for females aged 20–44 years at the time of the MEC examination	Nominal categorical	Control
BAXMSTAT	MEC examination status	Nominal categorical	Exclusion
MCQ160A	Whether a doctor or other health professional has ever told the participant that they had arthritis	Binary categorical	Exclusion
MCQ160C	Whether a doctor or other health professional has ever told the participant that they had coronary heart disease	Binary categorical	Exclusion
MCQ160E	Whether a doctor or other health professional has ever told the participant that they had a heart attack (myocardial infarction)	Binary categorical	Exclusion
MCQ160F	Whether a doctor or other health professional has ever told the participant that they had a stroke	Binary categorical	Exclusion
MCQ160M	Whether a doctor or other health professional has ever told the participant that they had another thyroid problem	Binary categorical	Exclusion
DR1TKCAL	Energy (kcal)	Continuous quantitative	Exclusion
DR1TSUGR	Total sugars (g)	Continuous quantitative	Exclusion
DR1TTFAT	Total fat (g)	Continuous quantitative	Exclusion
DR1TSODI	Sodium (mg)	Continuous quantitative	Exclusion
DR1TFIBE	Dietary fiber (g)	Continuous quantitative	Exclusion
SEQN	Respondent sequence number (unique participant identifier)	Identifier	Identification / Survey Design
WTINT2YR	Full-sample 2-year interview weight	Continuous quantitative	Identification / Survey Design

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Table 2 – continued from previous page

Variable Name	Variable Description	Measurement Scale	Data Classification
WTMEC2YR	Full-sample 2-year MEC examination weight	Continuous quantitative	Identification / Survey Design
WTSAF2YR	Fasting subsample 2-year MEC weight	Continuous quantitative	Identification / Survey Design
WTPH2YR	Phlebotomy 2-year weight	Continuous quantitative	Identification / Survey Design
SDMVPSU	Masked variance-unit pseudo-PSU variable for variance estimation	Nominal categorical	Identification / Survey Design
SDMVSTRA	Masked variance-unit pseudo-stratum variable for variance estimation	Nominal categorical	Identification / Survey Design

5 Data Quality Assessment and Data Cleaning

Before conducting statistical analyses, the dataset will undergo a systematic data quality assessment and cleaning process. This process will include the identification and evaluation of missing values, implausible measurements, special response codes, duplicate records, and logical inconsistencies.

Missing Data. Missing values will be identified and evaluated separately for each variable. No arbitrary values, such as zero, the mean, or the median, will be assigned to missing observations at this stage. Missing values will initially be retained as missing, and their patterns and potential implications for the analyses will be examined. Decisions regarding their treatment will be made at a later stage, considering the nature of missingness, the analytical requirements, and the appropriate NHANES sampling weights.

Special Response Codes. Variable-specific codes representing refusal, uncertainty, or unavailable responses will be identified using the NHANES documentation. These codes will be distinguished from valid measurements and response categories. For example, values 7 and 9 in SMQ020 will be treated as invalid responses for smoking classification and converted to missing values.

Duplicate Records. Participant identification numbers (SEQN) will be examined for duplicate records, particularly following the merging of different NHANES data files. The merging process will be checked to ensure that each participant is appropriately represented and that duplicate records are not unintentionally introduced.

Logical Inconsistencies. Relationships between variables will be evaluated according to NHANES questionnaire structure, eligibility criteria, and skip patterns. For example, the absence of SMQ040 among participants with SMQ020 = 2 will be interpreted as structurally missing information rather than an invalid response.

Analytical Treatment. Participants will not be automatically excluded from the master dataset because of missing information in individual variables. The treatment of missing observations and the inclusion criteria for each statistical analysis will be determined separately. The appropriate NHANES sampling weights will also be considered when defining analytical samples and conducting subsequent statistical analyses.

These procedures will aim to improve data reliability, preserve available information, minimize unnecessary data loss, and establish a consistent basis for subsequent descriptive and inferential analyses.

6 Frequency Tables and Empirical Distribution

6.1 Primary Variables, Physical Activity and Smoking Status

To provide an overview of the distribution of the categorical explanatory variables, frequency distributions were constructed for physical activity levels and smoking status. The distributions are presented using bar charts, showing both the number and percentage of participants in each category.

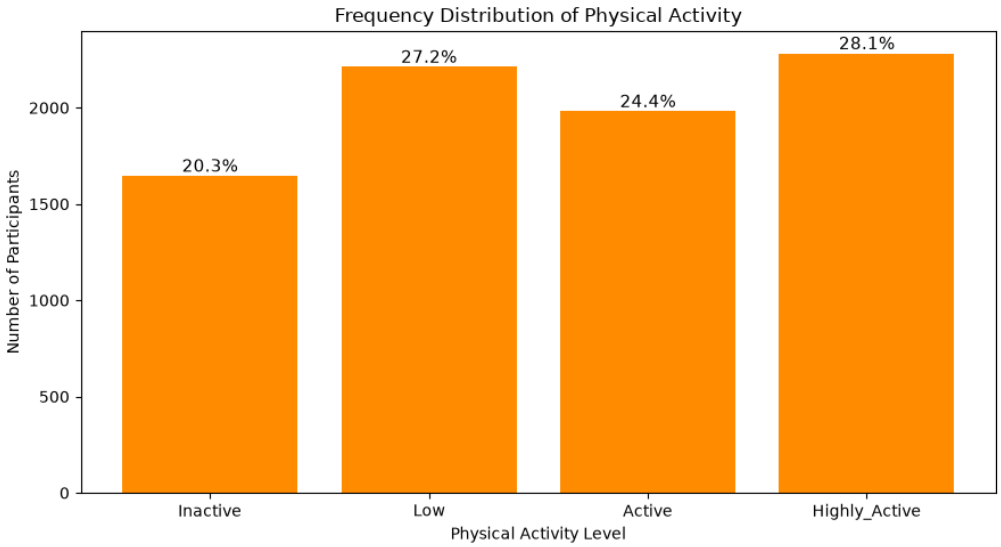


Figure 1: Frequency distribution of physical activity levels.

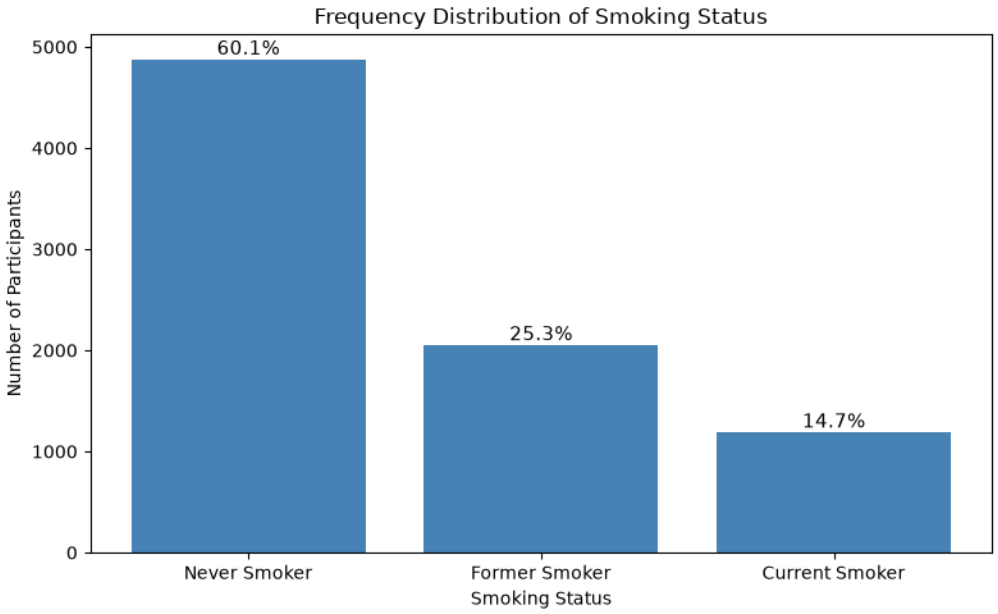


Figure 2: Frequency distribution of smoking status.

6.2 Metabolic Variables(BMI and Blood Pressure)

BMI The observed BMI distribution indicates substantial variation in body mass among the sampled adults. Although a large proportion of observations is concentrated between approximately 20 and 30 kg/m^2, the distribution also includes individuals with considerably higher BMI values. This suggests that the sample encompasses a wide range of body mass conditions, including individuals potentially exposed to increased metabolic and cardiovascular health risks associated with elevated BMI. Consequently, examining the relationship between BMI and lifestyle factors such as physical activity and smoking status may provide valuable insights into differences in body mass across population subgroups.

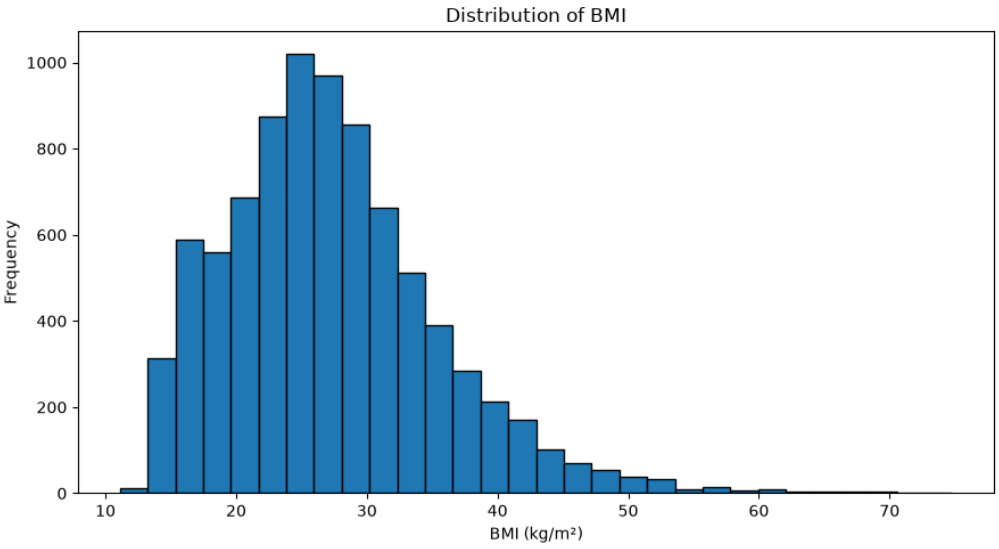


Figure 3: Empirical distribution of Body Mass Index (BMI).

Blood Pressure The empirical distribution of diastolic blood pressure is unimodal, with a concentration of observations between approximately 60 and 80 mmHg. The frequency gradually decreases toward higher values, producing a moderately extended right tail. This pattern indicates that most sampled adults exhibit diastolic blood pressure within a relatively concentrated range, while a smaller proportion have substantially higher readings. The presence of these higher values reflects variation in blood pressure levels across the sample and highlights the importance of examining potential associations between diastolic blood pressure and lifestyle factors. The empirical distribution of systolic blood pressure is unimodal and positively skewed, with most observations concentrated approximately between 100 and 130 mmHg. The frequency decreases progressively toward higher values, forming a relatively long right tail. This pattern indicates that although systolic blood pressure is concentrated within a central range for most sampled adults, a smaller proportion exhibits substantially higher readings. Such variation may reflect differences in cardiovascular health across participants and provides a relevant basis for investigating associations between systolic blood pressure and lifestyle-related factors.

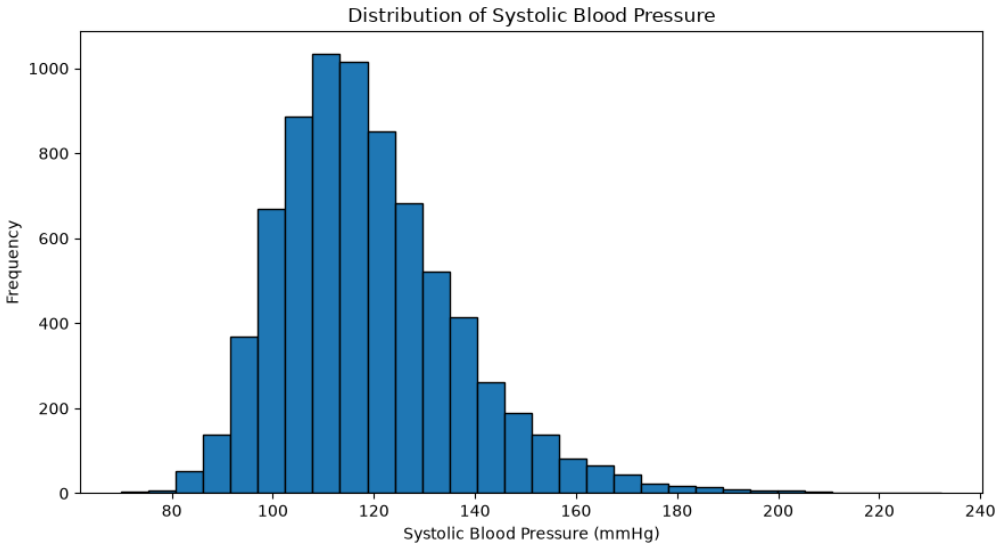


Figure 4: Empirical distribution of systolic blood pressure.

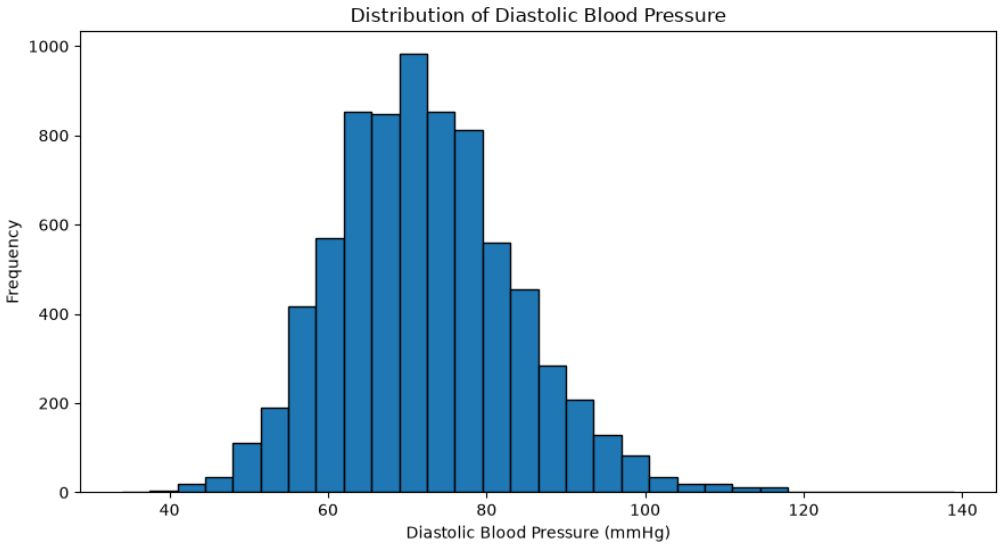


Figure 5: Empirical distribution of diastolic blood pressure.

6.3 Glycemic Biomarkers

Glycohemoglobin The empirical distribution of glycohemoglobin (HbA1c) is unimodal and strongly positively skewed, with a substantial concentration of observations between approximately 5 and 6 percent. The frequency decreases considerably as HbA1c increases, while a relatively long right tail extends toward higher values. This pattern suggests that most sampled adults exhibit HbA1c levels within a relatively concentrated range, whereas a smaller subgroup presents considerably elevated values. These higher measurements may reflect differences in glycemic regulation and the presence of individuals with less favorable metabolic profiles. Consequently, the distribution highlights the heterogeneity of glycemic status within the sample and supports further investigation of its association with lifestyle factors.

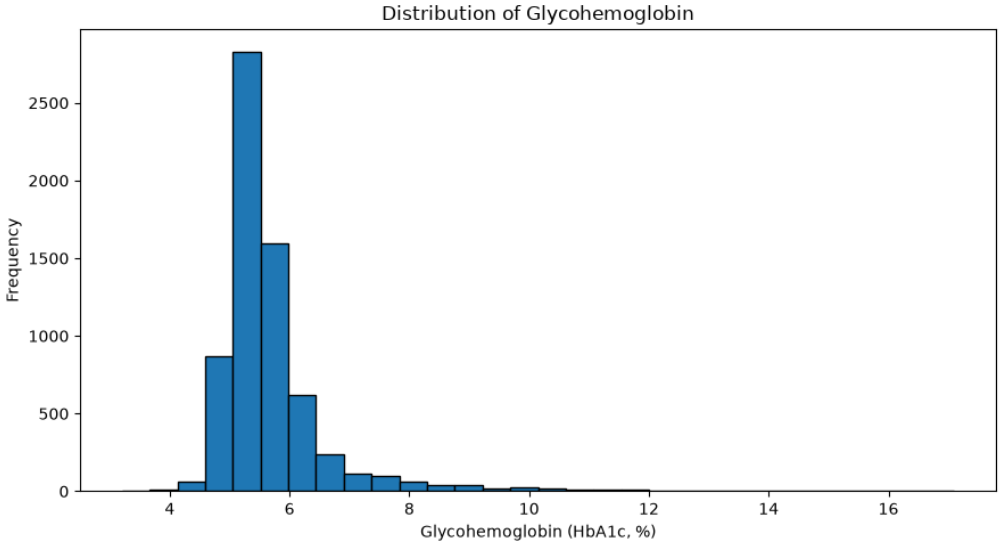


Figure 6: Empirical distribution of glycohemoglobin (HbA1c).

Fasting Glucose The empirical distribution of fasting glucose is unimodal and strongly positively skewed, with a pronounced concentration of observations around 100 mg/dL. Most measurements fall approximately within the range of 80120 mg/dL, while the frequency decreases substantially toward higher glucose levels. The extended right tail indicates that a smaller subgroup of participants exhibits considerably elevated fasting glucose concentrations. These higher values may reflect differences in glucose regulation and the presence of individuals with less favorable metabolic profiles. Overall, the distribution highlights heterogeneity in fasting glucose levels among sampled adults and provides a basis for investigating associations with lifestyle-related factors.

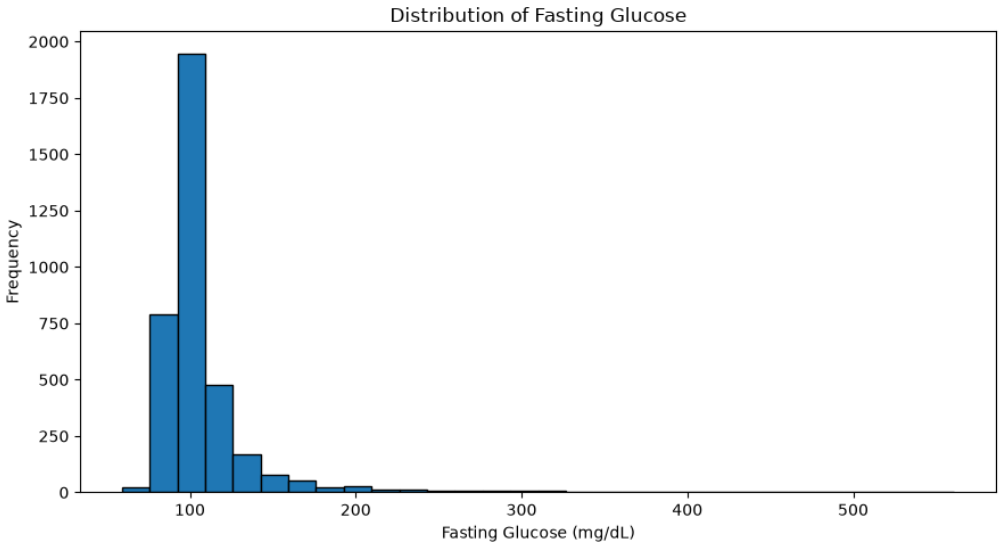


Figure 7: Empirical distribution of fasting glucose.

6.4 Lipid Profile

HDL-Cholesterol The empirical distribution of direct HDL-cholesterol is unimodal and moderately positively skewed, with a substantial concentration of observations between approximately 40 and 65 mg/dL. The highest frequencies occur around 4550 mg/dL, followed by a gradual decline toward higher concentrations. This pattern indicates that HDL-cholesterol levels are relatively concentrated among sampled adults, although meaningful variation exists across individuals. A smaller proportion of participants exhibits considerably higher HDL values, reflecting heterogeneity in lipid profiles. Given the role of HDL-cholesterol in cardiovascular health, this variation provides a relevant basis for examining its association with physical activity and smoking status.

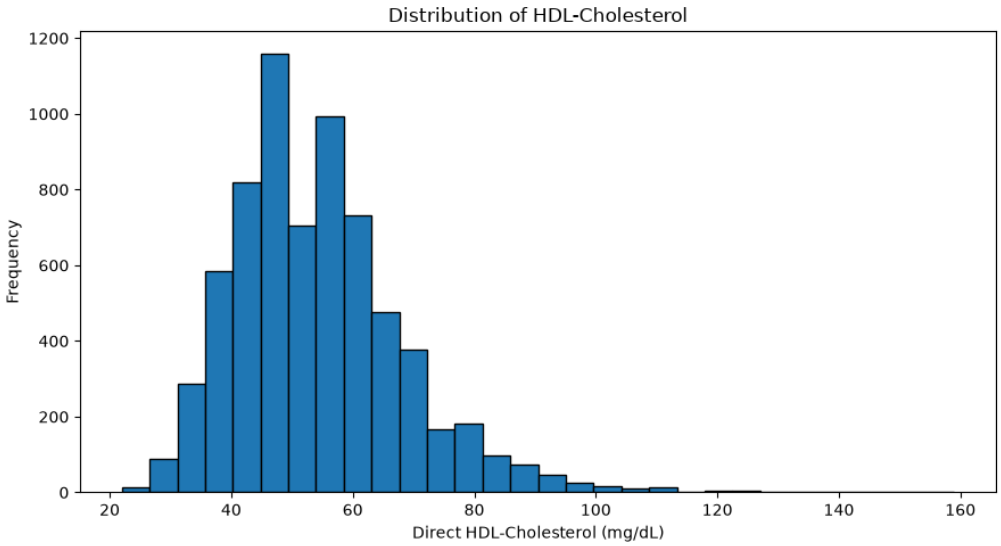


Figure 8: Empirical distribution of direct HDL-cholesterol.

Total Cholesterol The empirical distribution of total cholesterol is unimodal and moderately positively skewed, with a substantial concentration of observations between approximately 150 and 200 mg/dL. The highest frequencies occur around 175190 mg/dL, followed by a gradual decline toward higher concentrations. Although most sampled adults exhibit total cholesterol levels within a relatively concentrated range, a smaller subgroup presents considerably elevated values, including measurements above 240 mg/dL. This pattern reflects variation in lipid profiles across participants and highlights the potential relevance of elevated cholesterol levels to cardiovascular health. Consequently, examining associations between total cholesterol and lifestyle factors may provide useful insights into differences in lipid-related health outcomes.

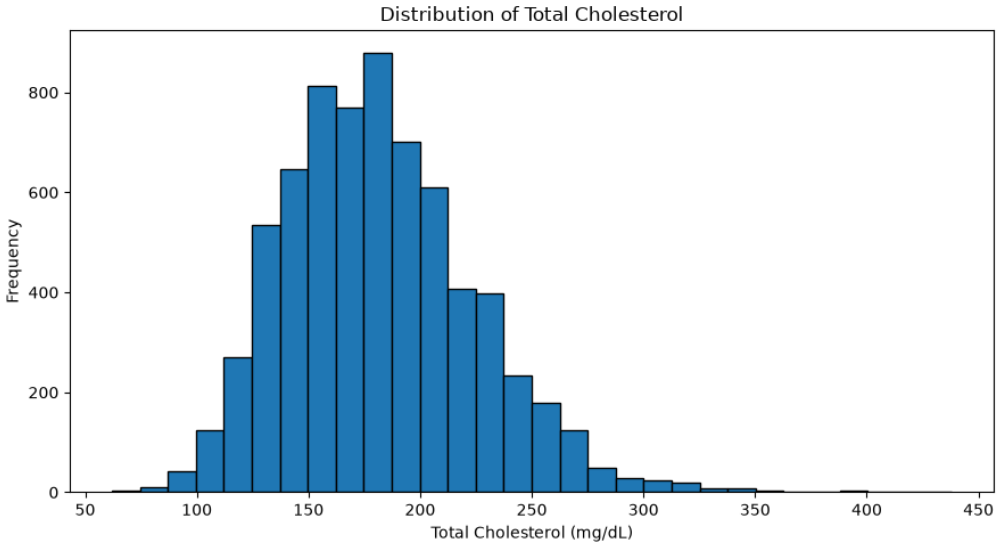


Figure 9: Empirical distribution of total cholesterol.

Triglycerides The empirical distribution of triglycerides is unimodal and strongly positively skewed, with a pronounced concentration of observations approximately between 50 and 150

mg/dL. The frequency decreases rapidly as triglyceride levels increase, while an exceptionally long right tail extends toward substantially higher values. This pattern indicates that most sampled adults exhibit relatively concentrated triglyceride levels, whereas a smaller subgroup presents markedly elevated concentrations. The presence of these high values reflects considerable heterogeneity in lipid metabolism across participants and may indicate less favorable metabolic profiles in a subset of the sample. This distribution highlights the relevance of investigating triglyceride levels in relation to physical activity and smoking status.

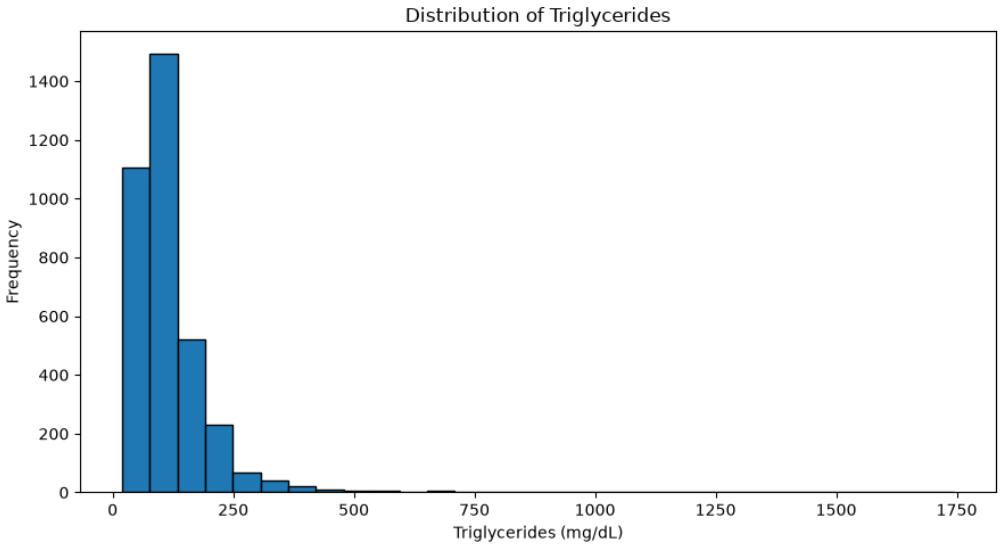


Figure 10: Empirical distribution of triglycerides.

7 Measures of Central Tendency and Dispersion

The measures of central tendency and dispersion were selected based on the empirical distribution of each response variable. The mean and standard deviation were considered appropriate for variables with approximately symmetric distributions, whereas the median and interquartile range (IQR) were preferred for variables exhibiting substantial positive skewness. The IQR was calculated as the difference between the third and first quartiles ($Q_3 - Q_1$), representing the spread of the central 50% of observations.

For Body Mass Index (BMXBMI), the distribution exhibited positive skewness. Therefore, the median and IQR were considered more representative measures of central tendency and dispersion, respectively, because they are less sensitive to unusually high BMI values.

Average systolic blood pressure (SYS_AVG) displayed a unimodal distribution with moderate positive skewness. The mean and standard deviation were selected as the primary descriptive measures to summarize the overall distribution, while the median and IQR were retained as complementary robust measures.

Average diastolic blood pressure (DIA_AVG) exhibited a relatively symmetric distribution with limited skewness. Consequently, the mean and standard deviation were considered suitable measures for describing its central tendency and variability.

Glycohemoglobin (LBXGH) and fasting glucose (LBXGLU) exhibited positively skewed distri-

butions, particularly fasting glucose, which contained a relatively extended upper tail. Accordingly, the median and IQR were preferred to summarize their central tendency and dispersion, as these measures are less influenced by extreme observations.

Direct HDL-cholesterol (LBDHDD) showed a unimodal distribution with relatively limited to moderate skewness. The mean and standard deviation were used to summarize its overall distribution, while the median and IQR provided complementary information about its central 50% of observations.

Total cholesterol (LBXTC) exhibited a comparatively symmetric distribution with some positive skewness. Therefore, the mean and standard deviation were selected as the primary descriptive measures.

Finally, triglycerides (LBXTLG) displayed substantial positive skewness and a pronounced upper tail. The median and IQR were therefore considered more appropriate measures for describing its central tendency and dispersion than the mean and standard deviation.

In addition to these measures, the minimum, maximum, and range were calculated for all response variables to describe their observed variability. However, the range was interpreted cautiously because it depends exclusively on the two extreme observations and is therefore particularly sensitive to outliers.

For completeness and comparability, the mean, median, standard deviation, first quartile, third quartile, interquartile range, minimum, maximum, and range were calculated for all response variables. The selection of preferred descriptive measures was guided by the observed distributional characteristics rather than applied uniformly across variables.

Table 3: Descriptive Statistics of Response Variables

Variable	Description	N	Missing	Mean	Median	SD	Q1	Q3	IQR	Min	Max	Range
BMXBMI	Body Mass Index (kg/m^2)	8471	389	27.25	26.40	8.14	21.60	31.70	10.10	11.1	74.8	63.7
SYS_AVG	Systolic Blood Pressure (mmHg)	7518	283	119.09	116.33	18.15	106.33	129.00	22.67	70.0	232.33	162.33
DIA_AVG	Diastolic Blood Pressure (mmHg)	7518	283	72.21	71.67	11.47	64.00	79.33	15.33	34.0	139.0	105.0
LBXGH	Glycohemoglobin (HbA1c, %)	6715	484	5.71	5.50	1.05	5.20	5.80	0.60	3.2	17.1	13.9
LBXGLU	Fasting Glucose (mg/dL)	3672	324	107.88	100.00	32.48	93.00	109.00	16.00	59.0	561.0	502.0
LBDHDD	Direct HDL-Cholesterol (mg/dL)	6890	1178	54.11	52.00	14.10	44.00	61.00	17.00	22.0	159.0	137.0
LBXTC	Total Cholesterol (mg/dL)	6890	1178	181.54	178.00	42.32	151.00	207.00	56.00	62.0	438.0	376.0
LBXTLG	Triglycerides (mg/dL)	3517	479	115.97	96.00	84.37	70.00	136.00	66.00	19.0	1745.0	1726.0

8 Descriptive Comparison of BMI and Glycohemoglobin Across Smoking Status and Physical Activity

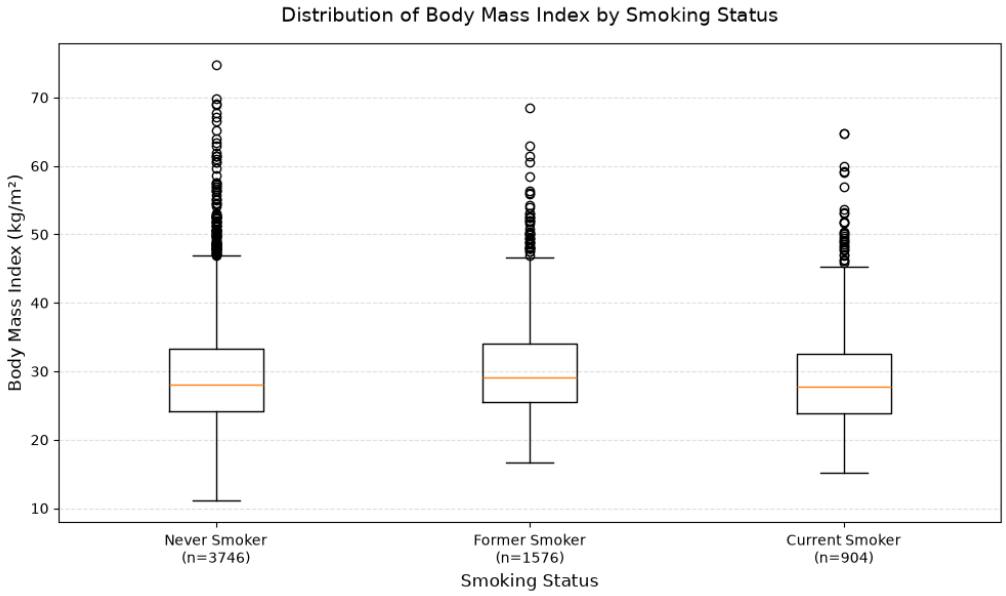


Figure 11: Distribution of Body Mass Index (BMI) across Smoking Status Categories among Adults Aged 18 Years and Older.

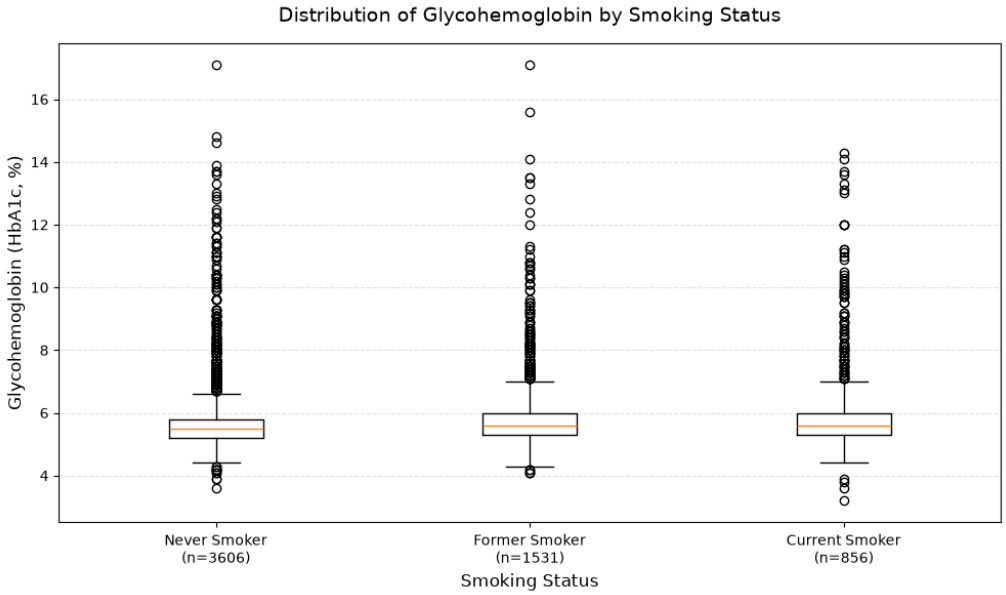


Figure 12: Distribution of Glycohemoglobin (HbA1c) across Smoking Status Categories among Adults Aged 18 Years and Older.

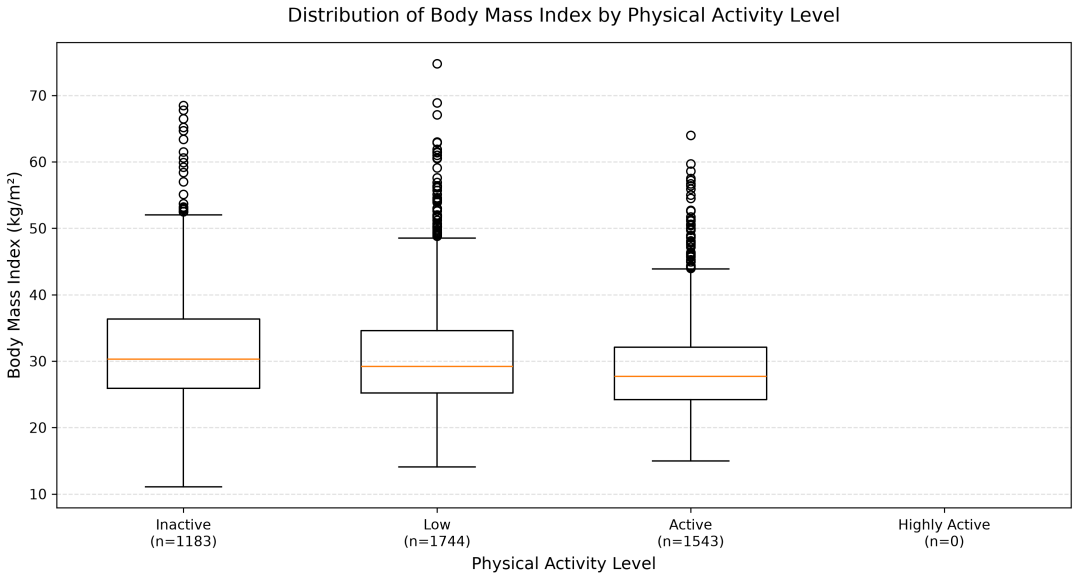


Figure 13: Distribution of Body Mass Index (BMI) by Physical Activity Level among Adults Aged 18 Years and Older.

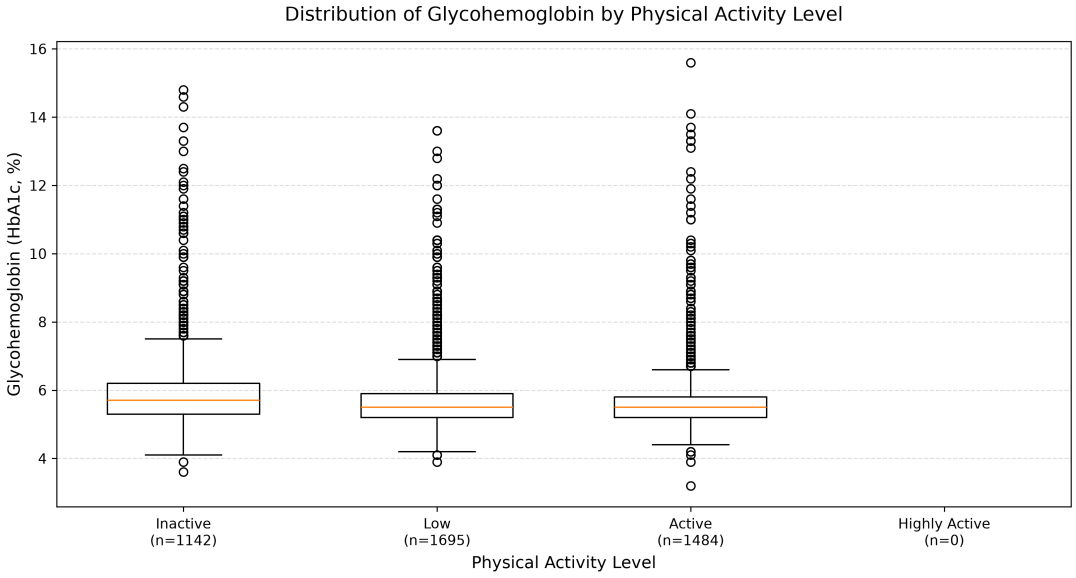


Figure 14: Distribution of Glycohemoglobin (HbA1c) by Physical Activity Level among Adults Aged 18 Years and Older.

9 Preliminary Descriptive Comparison

According to last part BoxPlot charts:

Smoking Status The distributions of body mass index (BMI) and glycohemoglobin (HbA1c) were compared across never, former, and current smokers using boxplots.

For BMI, former smokers exhibited the highest median, whereas current smokers showed a slightly lower median than never smokers. The interquartile ranges were relatively similar

across smoking categories, indicating comparable variability within the central 50 present of observations. All three groups displayed positively skewed distributions, with numerous upper-tail observations.

For HbA1c, the median values among former and current smokers appeared slightly higher than those among never smokers. However, the interquartile ranges were relatively similar across groups, and substantial overlap was observed between their distributions. Positive skewness and upper-tail observations were evident in all smoking categories.

Overall, the descriptive comparisons suggest modest differences in the central tendencies of BMI and HbA1c across smoking categories, while the distributions exhibit considerable overlap. These findings represent unadjusted descriptive comparisons and do not establish statistically significant differences or causal relationships between smoking status and either health outcome.

Physical Activity The distributions of Body Mass Index (BMI) and Glycohemoglobin (HbA1c) were compared across physical activity categories among adults aged 18 years and older.

For BMI, the median value appeared to decrease across the Inactive, Low, and Active groups, with approximate medians of 30, 29, and 27.5 kg/m^2, respectively. The Inactive group exhibited a relatively wider interquartile range, whereas the Active group showed a somewhat narrower central distribution. All three groups displayed right-skewed distributions with numerous upper outliers. Despite the observed differences in median BMI, considerable overlap between the distributions was evident.

For HbA1c, the Inactive group exhibited a slightly higher median (approximately 5.7percent) than the Low and Active groups (approximately 5.5 percent). The central distributions of the Low and Active groups were relatively similar, with substantial overlap between them. All three distributions showed right-skewness and numerous upper outliers.

Overall, the descriptive patterns suggest lower median BMI and slightly lower median HbA1c among participants classified as more physically active. However, the observed differences are relatively modest, and the substantial overlap between distributions indicates considerable variability within each activity category. These visual comparisons do not establish statistical significance or causal relationships.

No observations were available for the Highly Active category in either comparison; therefore, this group could not be evaluated.

10 Justification of the Final Statistical Question

The Final Statistical Question becomes: "How are cigarette smoking status and physical activity levels jointly associated with anthropometric, blood pressure, glycemic, and lipid indicators (BMI, systolic and diastolic blood pressure, Glycohemoglobin, Fasting Glucose, Direct HDL-Cholesterol, Total Cholesterol, and Triglycerides) among adults aged 18 years and older in the NHANES 20212023 population?"

The research question is appropriate for the NHANES 20212023 dataset because this survey provides information on cigarette smoking, leisure-time physical activity, anthropometric measurements, blood pressure, glycemic indicators, and lipid biomarkers within a nationally repre-

sentative sample of the U.S. civilian noninstitutionalized population.

The availability of these variables allows smoking status and physical activity levels to be examined jointly in relation to multiple health indicators. Furthermore, demographic information, including age, sex, and family income-to-poverty ratio, provides an opportunity to account for relevant potential confounding variables in subsequent statistical analyses.

The preliminary descriptive comparisons of BMI and HbA1c across smoking and physical activity categories demonstrated observable differences in their distributions, supporting the relevance of further investigation. Nevertheless, these initial comparisons are exploratory and do not establish statistical significance or causal relationships.

An additional strength of the dataset is the substantial number of observations available across the relevant smoking and physical activity categories. The preliminary data assessment indicated that several of these subgroups contained considerable numbers of participants, providing a reasonable basis for descriptive comparisons and further statistical investigation. Although the effective sample size varies across health outcomes due to missing observations, the available data offer meaningful opportunities to examine associations between lifestyle characteristics and health indicators. Nevertheless, subgroup sample sizes and data availability should be evaluated individually before conducting statistical analyses.

Although the dataset is suitable for addressing the research question, several limitations must be considered. Missing observations may reduce the effective sample size for individual outcomes, smoking and physical activity information may be affected by self-reporting errors, and the complex survey design requires appropriate consideration of sampling weights, strata, and primary sampling units.

11 Requirements Parts

11.1 Identification of Files and Merge Key

The NHANES August 2021–August 2023 dataset was used as the primary source of information. Multiple component files were combined to construct an analytical dataset containing demographic characteristics, lifestyle variables, anthropometric measurements, blood pressure measurements, and laboratory biomarkers.

The following files were used:

- `DEMO_L`: Demographic information, including age, sex, income-to-poverty ratio, survey weights, strata, and primary sampling units.
- `BMX_L`: Body measurements, including Body Mass Index (BMXBMI).
- `BPXO_L`: Oscillometric blood pressure measurements.
- `GHB_L`: Glycohemoglobin (LBXGH).
- `GLU_L`: Fasting glucose (LBXGLU).
- `HDL_L`: Direct HDL-Cholesterol (LBDHDD).

- `TCHOL_L`: Total Cholesterol (LBXTC).
- `TRIGLY_L`: Triglycerides (LBXTLG).
- `PAQ_L`: Physical activity questionnaire data.
- `SMQ_L`: Cigarette smoking questionnaire data.

The respondent sequence number (`SEQN`) was used as the common merge key. This identifier uniquely identifies individual respondents within the NHANES dataset and enables participant-level linkage across the different survey components.

The files were merged using participant identifiers rather than row positions. This approach ensures that measurements from different components are assigned to the correct individuals. The demographic file was used as the initial reference dataset, and the other component files were subsequently linked using left joins, preserving the demographic records and allowing component-specific missing observations to remain identifiable.

Table 4 presents the number of participants remaining after matching the physical activity and smoking status datasets with each health outcome variable.

Table 4: Sample Size After Merging with Each Health Variable

Health Variable	**After Physical Activity Merge (n)**	**After Smoking Merge (n)**
BPXOSY1	5,673	6,947
BPXOSY2	5,667	6,940
BPXOSY3	5,652	6,923
BPXODI1	5,673	6,947
BPXODI2	5,667	6,940
BPXODI3	5,652	6,923
BMXBMI	5,530	6,306
LBXGH	5,561	6,715
LBXGLU	3,068	3,672
LBDHDD	5,291	6,391
LBXTC	5,291	6,391
LBXTLG	2,935	3,517

11.2 Identification of Survey Weights and Sampling Design

NHANES employs a complex, multistage probability sampling design. Consequently, participants do not necessarily represent the same number of individuals in the target population. Each participant is assigned a sampling weight that reflects their probability of selection and adjustments for survey nonresponse and population calibration.

The appropriate sampling weight depends on the survey component and the variables included in a particular analysis. The following weights were identified for the present study:

- `WTMEC2YR`: Mobile Examination Center (MEC) examination weight, appropriate for analyses involving anthropometric measurements, blood pressure, and laboratory variables measured in the MEC.
- `WTSAF2YR`: Fasting subsample weight, appropriate for fasting laboratory measurements, including fasting glucose and triglycerides.
- `WTPH2YR`: Phlebotomy weight, used for laboratory analyses such as glycohemoglobin when specified by the component documentation.
- `WTINT2YR`: Interview weight, relevant when analyses are restricted to variables collected during the household interview.

For the present study, the weight corresponding to the most restrictive measurement component included in each outcome-specific analysis was selected. In particular, fasting glucose and triglycerides require fasting subsample weights, whereas BMI, blood pressure, HDL-Cholesterol, and Total Cholesterol are analyzed using the MEC examination weight. Glycohemoglobin is analyzed using the phlebotomy weight.

The sampling design variables `SDMVSTRA` and `SDMVPSU` were retained to identify the survey strata and primary sampling units, respectively. These variables are required to account for the complex sampling structure when estimating design-based standard errors and confidence intervals.

The sampling weights are not ordinary frequency weights. They represent the number of population members associated with each sampled participant under the NHANES sampling and adjustment procedures. Therefore, unweighted descriptive summaries and weighted population estimates may differ substantially.

The selection of the appropriate weight is outcome-specific and follows the NHANES analytic guidance. The weights were not used interchangeably across analyses.

11.3 Comparison of Unweighted and Weighted Summaries

Body Mass Index (BMI) For adults aged 18 years and older, the unweighted and weighted mean BMI values were calculated using the NHANES 2021–2023 examination data. The analysis included 6,235 participants with valid BMI measurements and positive MEC examination weights.

Table 5: Unweighted and Weighted BMI Summary

Statistic	**Value**
Sample Size	6,235
Unweighted Mean BMI (kg/m^2)	29.659
Weighted Mean BMI (kg/m^2)	29.542
Difference (Weighted – Unweighted)	-0.118

The unweighted mean BMI among adult participants was 29.659 kg/m^2, whereas the weighted mean was 29.542 kg/m^2. The difference between these estimates was approximately -0.118

kg/m^2, indicating that applying the NHANES examination weights resulted in a slightly lower estimated population mean BMI.

The relatively small difference between the two estimates suggests that, for BMI, the weighting procedure had a limited effect on the estimated mean compared with the unweighted sample mean. Nevertheless, the weighted estimate is more appropriate for describing the target population because NHANES employs a complex sampling design rather than simple random sampling.

This comparison illustrates the importance of accounting for sampling weights when estimating population-level health characteristics. However, the similarity between the two means does not imply that the sample is fully representative of the US adult population, nor does it establish that weighting has negligible effects on other statistical estimates, such as variances, standard errors, or subgroup comparisons.

Furthermore, this comparison concerns BMI only. Differences between weighted and unweighted estimates may vary across other health outcomes and demographic subgroups, particularly when the corresponding variables are measured in specialized subsamples.

11.4 Definition of the Analytical Population

The initial NHANES 2021–2023 demographic dataset included 11,933 participants. The analytical population was restricted to adults aged 18 years and older, corresponding to the eligibility criteria of the present research question.

Participants were retained in the master dataset regardless of whether they had complete observations for every study variable. Missing values were handled separately for each outcome-specific analysis.

The following inclusion and exclusion criteria were applied:

- **Age restriction:** Participants younger than 18 years were excluded from the analytical population.
- **Missing explanatory variables:** Participants without valid smoking status or physical activity classification were excluded from analyses requiring these variables.
- **Missing outcome measurements:** Participants without an observed value for the specific outcome were excluded from that outcome-specific analysis.
- **Missing covariates:** Participants without complete observations for the covariates included in a particular statistical model were excluded from that model using complete-case analysis.

After applying the age restriction, 8,153 participants remained in the adult population. The number of observations available for subsequent analyses varied according to the completeness of the smoking, physical activity, outcome, and covariate variables.